

EX:7 WRITE THE PROGRAM FOR BREADTH-FIRST SEARCH (BFS)

AIM

To develop a Python program that implements the **Breadth-First Search (BFS)** algorithm to traverse a graph level by level from a given starting node.

ALGORITHM

Step 1: Start the program.

Step 2: Create the graph using an adjacency list.

Step 3: Accept the starting node from the user.

Step 4: Mark the starting node as visited and insert it into a queue.

Step 5: Remove a node from the queue, display it, and visit all its unvisited adjacent nodes by adding them to the queue.

Step 6: Repeat the process until the queue becomes empty, then end the program.

PROGRAM CODE:

```
from collections import deque
```

```
graph = {}
```

```
n = int(input("Enter number of vertices: "))
```

```
for i in range(n):
```

```
    vertex = input("Enter vertex: ").strip()
```

```
    neighbours = input("Enter neighbours separated by space: ").split()
```

```
    graph[vertex] = neighbours
```

```
start = input("Enter starting node: ").strip()
```

```

visited = set()
queue = deque([start])

print("\nBFS Traversal:")

while queue:
    vertex = queue.popleft()

    if vertex not in visited:
        print(vertex, end=" ")
        visited.add(vertex)

    for neighbour in graph.get(vertex, []):
        if neighbour not in visited:
            queue.append(neighbour)

```

OUTPUT:

```

===== RESTART: C:/Users/Hp/AppData/Local/Programs/Python/Python
Enter number of vertices: 4
Enter vertex: A
Enter neighbours separated by space: B C
Enter vertex: B
Enter neighbours separated by space: A D
Enter vertex: C
Enter neighbours separated by space: A
Enter vertex: D
Enter neighbours separated by space: B
Enter starting node: A

BFS Traversal:
A B C D

```

RESULT:

The Python program successfully implements the Breadth-First Search (BFS) algorithm to traverse the graph. The nodes are visited and displayed in **level-by-level (breadth-first)** order starting from the given source node.